

Section 1

Game Theory

Introduction and Examples

Games against nature (decision theory for a single agent)

Expected utility theory for a single agent is sometimes called the theory of "games against nature". Consider this example.

Example 1: Planning a party: Our agent is planning a party, and is worried about whether it will rain or not. The utilities and probabilities for each state and action can be represented as follows:

Normal or Matrix from		Nature's states	
		Rain ($p=1/3$)	No rain ($\sim p=2/3$)
Party planner's possible actions	Outside	1	3
	Inside	2	2

https://web.stanford.edu/class/symsys202/Game_Theory_Through_Examples.html

Expected Utility

The expected utility of an action A given uncertainty about a state S

$$EU(A|S) = \text{Prob}(S|A) * \text{Utility}(S|A) + \text{Prob}(\text{not } S|A) * \text{Utility}(\text{not } S|A)$$

(Prob. = Probability)

- Note that action A can be viewed as a compound gamble or outcome.
- Also, note that the probability of a state can depend on the agent's choice of action, although, in the above example, it does not.

Expected Utility

$$EU(A) = \text{Prob}(S|A) * \text{Utility}(S|A) + \text{Prob}(\text{not } S|A) * \text{Utility}(\text{not } S|A)$$

For the party problem:

$$EU(\text{Outside}) = (1/3)(1) + (2/3)(3) = 2.33;$$

$$EU(\text{Inside}) = (1/3)(2) + (2/3)(2) = 2$$

Therefore, choose Outside, the action with the higher expected utility.

(Noncooperative) game theory (decision theory for more than one agent, each acting autonomously (no binding agreements))

In the examples below, we'll assume two self-utility maximizing agents (or players), each of whom has complete information about the options available to themselves and the other player as well as their own and the other's payoffs (utilities) under each option.

Example 2: Friends hoping to see each other

Consider two people, Chris and Kim. They both enjoy each other's company, but neither can communicate with the other before deciding whether to stay at home (where they would not see each other) or go to the beach this afternoon (where they could see each other). Each prefers going to the beach to being at home, and prefers being with the other person rather than being apart. This game can be represented by the following normal (or matrix) form:

Normal or Matrix form		Kim	
		Home	Beach
Chris	Home	(0,0)	(0,1)
	Beach	(1,0)	(2,2)

Strategy, Outcome, Payoff

Each player has a set of **strategies** ($=\{ \text{Home, Beach} \}$ for both players in this example).

Specifying one strategy i for the row player (Chris) and
one strategy j for the column player (Kim)

yields an **outcome**, which is represented as a pair of **payoffs** (R_{ij}, C_{ij}) ,
where R_{ij} is the utility the row player receives,
and C_{ij} is the utility the column player receives.

Dominant Strategy

In this example, going to the beach is a (strictly) **dominant strategy** for each player, because it always yields the best outcome, no matter what the other player does. Thus, if the players are both maximizing their individual expected utilities, each will go to the beach. So Beach-Beach is a dominant strategy equilibrium for this game. Because of this, Kim and Chris, if they are rational, do not need to **cooperate** (make an agreement) ahead of time. Each can just pursue their own interest, and the best outcome will occur for both.

(Noncooperative) game theory (decision theory for more than one agent, each acting autonomously (no binding agreements))

Example 3 - "Friends" with asymmetric preferences

Now consider Betty and John. John likes Betty, but Betty doesn't like John that much. Each knows this, and neither wants to call the other before deciding what to do this afternoon: stay at their respective homes or go to the neighborhood swimming pool. Here is the normal form:

Normal or Matrix form		John	
		Home	Pool
Betty	Home	(2,0)	(2,1)
	Pool	(3,0)	(1,2)

Dominant Strategy

In this case, Betty's best strategy depends on what John does. But if she assumes John is rational, she will reason that he will not stay home, because going to the pool is **a dominant strategy** for him. Knowing this, she can decide to stay home (because $2 > 1$). This is called **iterated dominance**. In this example, Betty gets higher utility than John because of their relative preferences, and John gets less utility than he would have if Betty wanted to be with him.

Pareto Optimal (Efficient)

In this example, Pool-Home (3,0), Home-Pool (2,1), and Pool-Pool (1,2) are all **Pareto optimal** outcomes. An outcome is **Pareto optimal** (or **efficient**) if no agent can be made better off than that outcome without making another agent worse off. The equilibrium outcomes in both this example and the previous one are Pareto optimal.

(Noncooperative) game theory (decision theory for more than one agent, each acting autonomously (no binding agreements))

Example 4 - Prisoners' dilemma

Consider Stan and Leland, two prisoners who have each been offered a deal to turn state's witness (defect) against the other. They can't communicate. They had originally agreed to remain in solidarity, i.e. not testify against each other, but since the agreement cannot be enforced, each must choose whether to honor it. If both remain in solidarity, then they will each only be convicted of a minor charge. If only one defects, then the state will throw the book at the other and let the defector go. If they both defect, each will get convicted of a serious charge. The payoff matrix (higher positive utility implies a better outcome) is as follows:

Normal or Matrix from		Leland	
		Solidarity	Defection
Stan	Solidarity	(3,3)	(1,4)
	Defection	(4,1)	(1,1)

Dominant Strategy (weakly, strictly)

In this game, the strategy of defection is **weakly dominant** for each player, meaning that whatever the other player does, defecting yields an outcome at least as good and possibly better than remaining in solidarity would. Note that if the bottom right cell payoffs were (2,2) instead of (1,1), then defecting would be **strictly dominant** for each player. Either way, Defection-Defection is a **dominant strategy** equilibrium. However, it is not **Pareto optimal**. Both players could be made better off if neither defected against the other.

Social Dilemma

This is an example of a **social dilemma**: a situation in which each agent's autonomous maximization of self-utility leads to an inefficient outcome. Such a situation can occur for any number of people, not just two. An agreement by two people to trade with each other (involving goods, services, and/or money) set's up a prisoners' dilemma-type game whenever the agreement cannot be enforced.

Coordination

Example 5 - Coordination

Let's go back to Chris and Kim. They are going to the same conference, and each is expecting the other to be there, but they haven't seen each other yet. The conferees have their choice of two activities on the first afternoon: swimming or hiking. They both hope to see each other -- if they don't they will have no fun, and each prefers swimming over hiking. They must each decide what to do before knowing where the other is going. Here is the normal form:

Normal or Matrix form		Kim	
		Swim	Hike
Chris	Swim	(2,2)	(0,0)
	Hike	(0,0)	(1,1)

Nash Equilibrium

The best outcome is obviously Swim-Swim, but going swimming is not dominant for either player. Both Swim-Swim and Hike-Hike have the property that each player's strategy is the best (or tied for the best) response to the other player's strategy in that pairing. This defines a more general equilibrium notion called the **Nash equilibrium**. The dominance equilibria of examples 1-3 are all Nash equilibria as well.

Mixed strategy

A third equilibrium exists in this game involving what are called **mixed strategies**. A **mixed strategy** is a probability distribution over the pure strategies (which are Swim and Hike for each player in this example). (Note that the players do not have to have the same set of strategies available to them, even though that has been the case in all our examples.)

In this example, if each player individually throws a die and goes swimming if the die comes up 1 or 2, and goes hiking if the die comes up 3, 4, 5, or 6, the resulting expected utility ($2/3$ for each player) cannot be improved upon for either player given that the other player uses this strategy.

Mixed strategy

In 1950, **John Nash** (depicted somewhat fictitiously in the film *A Beautiful Mind* -- the book is more accurate!) proved that every **finite game**, involving any number of players, has **at least one (Nash) equilibrium**, though there might not be any that involve only pure strategies for all players. In this example, there are three equilibria: the mixed strategy equilibrium (Swim, $1/3$; Hike, $2/3$)-(Swim, $1/3$; Hike, $2/3$), and two pure strategy equilibria -- Swim-Swim and Hike-Hike.

Theorem 1.1 (Nash's Theorem) Every finite two-player game has a Nash equilibrium.

Equilibrium Selection

When there is more than one equilibrium, and players cannot make binding agreements, they must try to coordinate to arrive at an equilibrium outcome. When only one equilibrium is also Pareto optimal, as Swim-Swim is in this case, that fact should suggest to rational players that it will be the one around which they coordinate. Many other criteria for equilibrium selection have been studied (e.g. **focal points**, **subgame perfection**, **stability** -- see the reading on game theory).

Coordination

Example 6 - "Battle of the sexes"

Finally, let's consider Roy and Jen. They are going to the same conference as Kim and Chris in example 5. They each would prefer to be in the same place (the swim or the hike), but their preferences differ about which it should be. Roy would rather go swimming, and Jen would rather go hiking. Here is the matrix form:

Normal or Matrix form		Jen	
		Swim	Hike
Roy	Swim	(3,2)	(1,1)
	Hike	(1,1)	(2,3)

Nash Equilibrium

This game has three Nash equilibria: Swim-Swim, Hike-Hike, and $(\text{Swim}, 2/3; \text{Hike}, 1/3) - (\text{Swim}, 1/3; \text{Hike}, 2/3)$.

Note that the mixed strategies differ for each player in the third equilibrium: each goes to their preferred activity with $2/3$ probability.

Focal Point

All of the equilibria are Pareto optimal this time, so that does not help for selection. Only the mixed strategy equilibrium results in equal expected utilities for the two players, so if both value equality or symmetry, this might be the **focal point**. But of course it will be difficult for Roy and Jen to see that unless **they have studied game theory!**

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Zero-Sum Games and Minimax Theorem

Imagine you are playing a game against a *super-smart* opponent. You want to find the move that gives you the highest possible reward (*greedily*). But you know your opponent is also smart, and they will try to block your best move and force you into a bad position.

If you play greedily, aiming for the best possible outcome, a smart opponent will trap you, and you will end up with the *worst possible outcome*.

Zero-Sum Games and Minimax Theorem

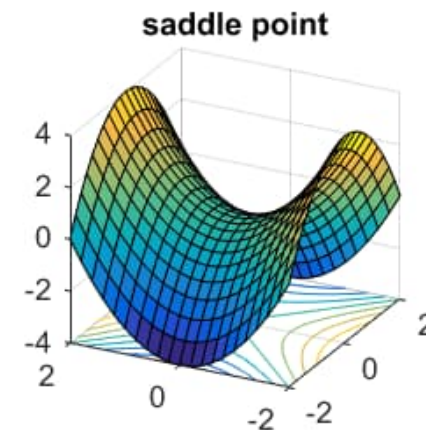
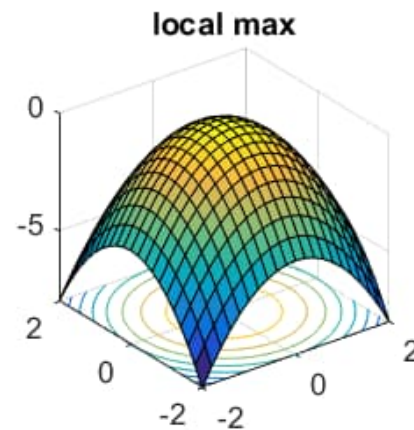
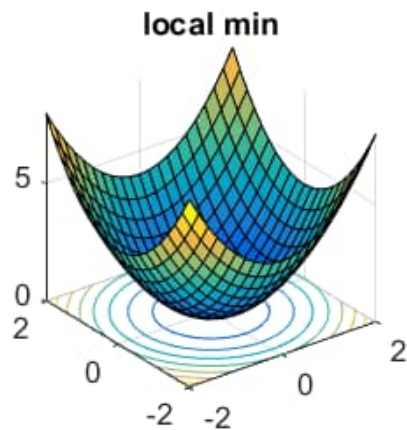
The Solution: “*Maximize your Minimum*”

Von Neumann (1928) realized that in a zero-sum game, you shouldn't look for the best-case scenario. You should look for the best of the worst-case scenarios.

- *Minimax* means: You want to minimize your maximum possible loss.
- (Alternatively called *Maximin*): You want to maximize your minimum possible gain.

Zero-Sum Games and Minimax Theorem

Saddle-point dynamics:

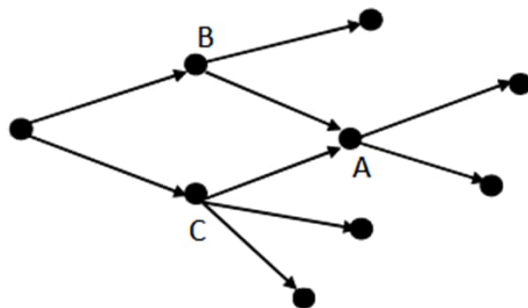


Game Tree

Game Tree: A tree is a set of nodes and directed edges connecting these nodes such that

- 1. there is an initial node, for which there is no incoming edge;
- 2. for every other node, there is exactly one incoming edge;
- 3. for any two nodes, there is a unique path that connect these two nodes.

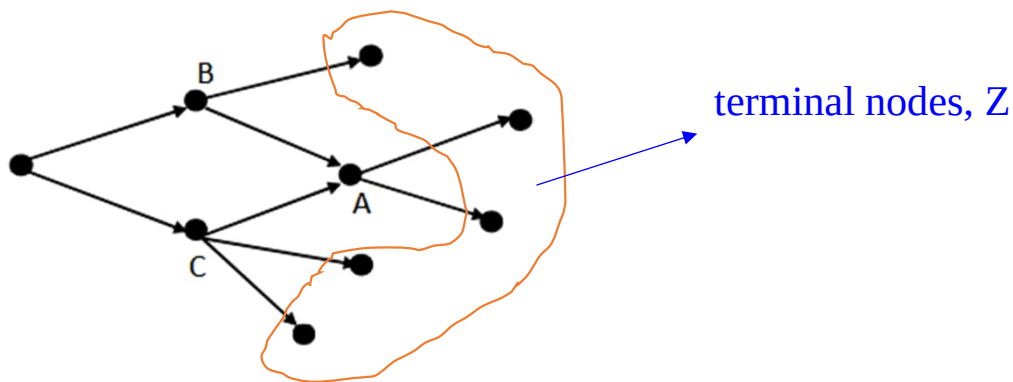
The nodes that are not followed by another node are called terminal



Games in Extensive-form

Definition (*Extensive form*) A Game consist of

- a set of players, $N = \{1, 2, \dots, n\}$, generic players $i, j \in N$
- a tree,
- an allocation of non-terminal nodes of the tree to the players,
- an *informational partition* of the non-terminal nodes, and
- payoffs for each player at each terminal node.



Outcome and Payoffs
(At each terminal node)

$$u_i : Z \rightarrow \mathbb{R}$$

Games in Extensive-form

Example) Matching Pennies with Perfect Information

These payoffs are von Neumann-Morgenstern utilities. That is, each player tries to maximize the expected value of his own payoffs given his beliefs about how the other players will play the game.

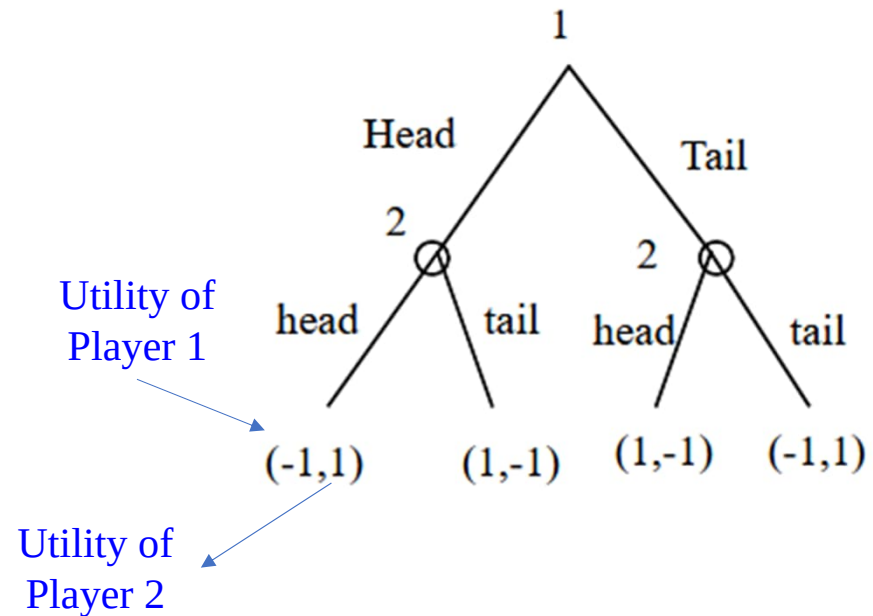


Figure 3.4: Matching Pennies with Perfect Information

Information Set

Definition An *information set* is a collection of nodes such that

1. the **same player i** is to move at each of these nodes;
2. the **same moves** are available at each of these nodes.

Definition An *information partition* is an allocation of each non-terminal node of the tree to an information set; the starting node must be “alone”.

A game is said to have *perfect information* if every information set has only one element.

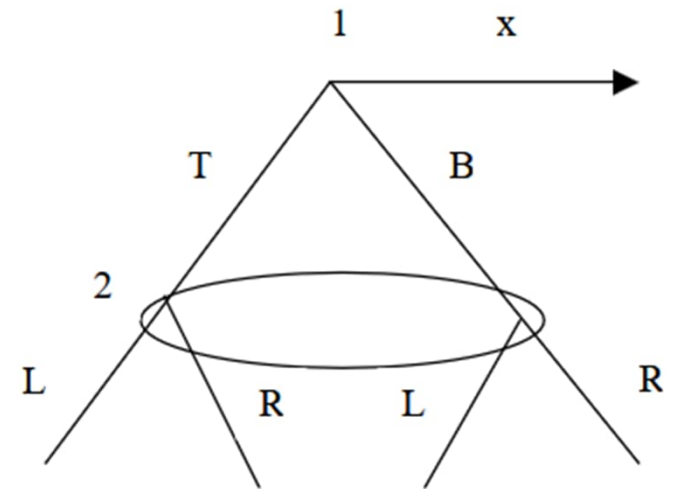


Figure 3.5: A game

Perfect Information

Definition A game is said to have *perfect information* if every information set has only one element.

- Recall that in a tree, each node is reached through a unique path. Hence, in a perfect-information game, a player can construct the previous play perfectly.
- For instance in Figure, Player 2 knows whether Player 1 chose Head or Tail. And Player 1 knows that when he plays Head or Tail, Player 2 will know what Player 1 has played.

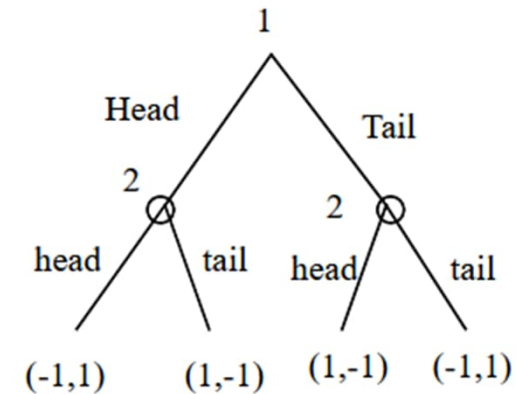
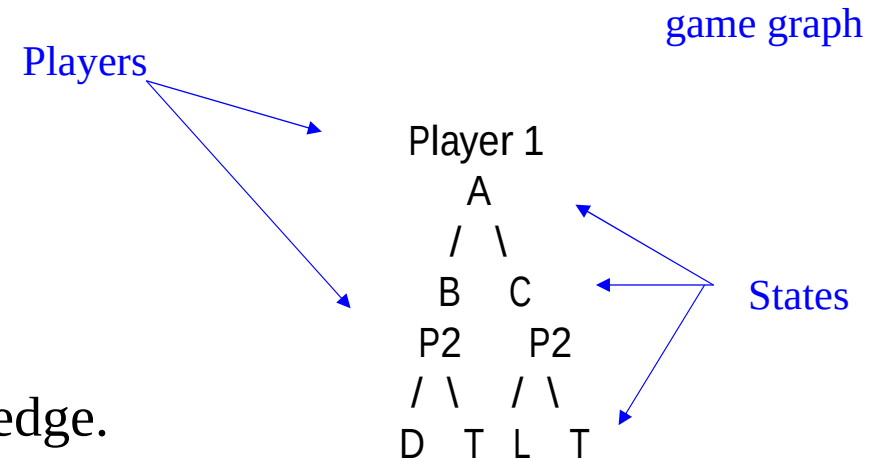


Figure 3.4: Matching Pennies with Perfect Information

Game Graphs and Reachability Games

Suppose:

- Player 1 wants to reach T.
- Player 2 wants to prevent reaching T.
- The token starts at A.
- The owner of a node chooses the outgoing edge.



Can Player 1 force the token to reach T?

- its solution can be explained through backward propagation / attractors.

Actions: {Left, Right}

Strategy: A seq. Of Actions: {LR}

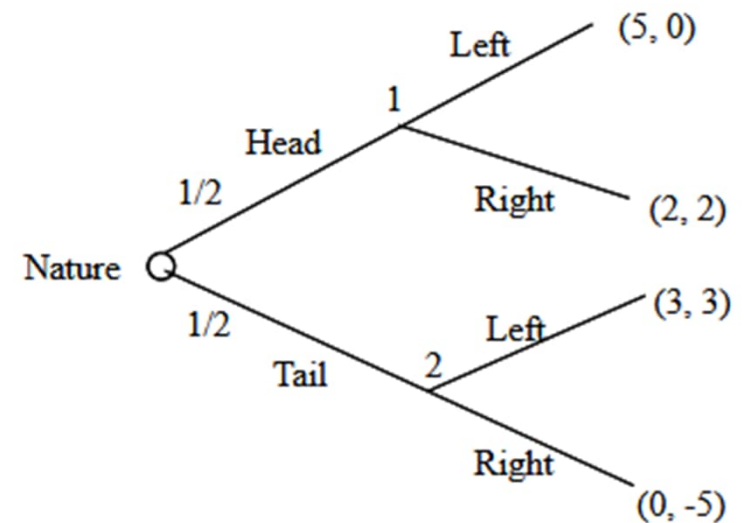
compute the region from which Player 1 can force the target and the complementary region from which Player 2 can prevent it.

Nature as a player

A Game with chance.

Definition A strategy of a player is a complete contingent-plan determining which action he will make at each information set he is to move (including the information sets that will not be reached according to this strategy).

More mathematically, a strategy of a player i is a function s_i that maps every information set h_i of player i to an action that is available at h_i .



Strategy $s_i: h_i \rightarrow a_i$

Strategies

Player 1 has only one information set. Hence, the set of strategies for Player 1 is {Head, Tail}.

On the other hand, Player 2 has two information sets. Hence, a strategy of Player 2 determines what to do at each information set, i.e., depending on what Player 1 does.

So, her strategies are:

HH = Head if Player 1 plays Head, and Head if Player 1 plays Tail

HT = Head if Player 1 plays Head, and Tail if Player 1 plays Tail;

TH = Tail if Player 1 plays Head, and Head if Player 1 plays Tail;

TT = Tail if Player 1 plays Head, and Tail if Player 1 plays Tail.

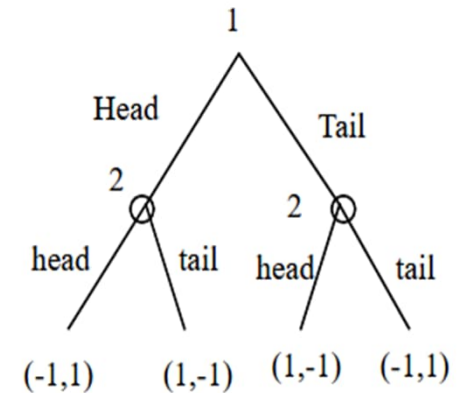
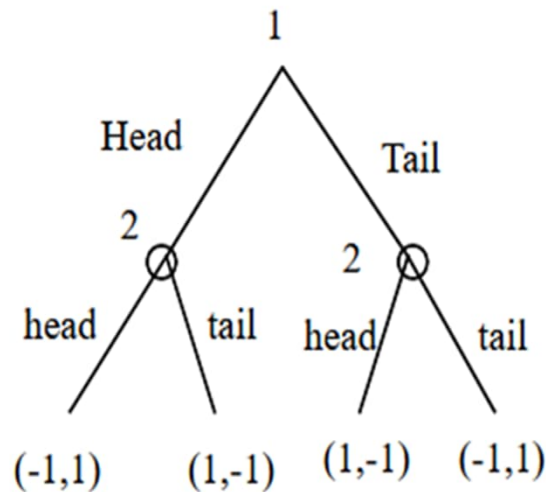
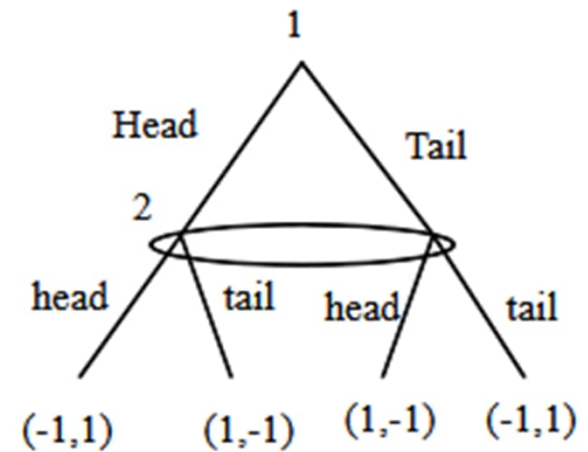


Figure 3.4: Matching Pennies with Perfect Information

Normal-form vs Extensive-form



	<i>HH</i>	<i>HT</i>	<i>TH</i>	<i>TT</i>
<i>Head</i>	$-1, 1$	$-1, 1$	$1, -1$	$1, -1$
<i>Tail</i>	$1, -1$	$-1, 1$	$1, -1$	$-1, 1$



$1 \backslash 2$	<i>head</i>	<i>tail</i>
<i>Head</i>	$-1, 1$	$1, -1$
<i>Tail</i>	$1, -1$	$-1, 1$

Types of Game (Summary)

1. Zero-sum vs. non-zero-sum games
2. Symmetric vs. asymmetric games (Sym: equal rules, strategic choices)
3. Simultaneous move vs. sequential move games
4. Cooperative vs. non-cooperative games
5. Perfect vs. imperfect information

(Perfect information: full transparency, every move is visible)